

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The scarcity of expert annotation in specialized domains such as medical imaging remains a fundamental obstacle to training robust vision models. Masked image modeling (MIM) offers a compelling self-supervised remedy, yet prevailing formulations rely on random, high-ratio masking that is at once computationally inefficient and poorly aligned with the semantics of downstream tasks; region-aware variants, in turn, inherit reconstruction heuristics or supervised signals that constrain their adaptability across tasks and modalities. We introduce Mask What Matters (MWM), a controllable, text-guided masking framework for self-supervised medical image analysis that resolves this tension by recasting an open-vocabulary prompt as a masking prior. A frozen vision-language model converts a natural-language description into zero-shot localization cues, from which MWM derives task-relevant regions and applies differentiated, region-aware masking—concentrating the reconstruction signal on diagnostically salient structures while suppressing the redundant background that dominates medical images. Because the masking distribution is conditioned directly on downstream semantics rather than on architecture or reconstruction difficulty, pretraining and transfer become explicitly coupled, yielding sharper semantic alignment and more transferable representations without recourse to per-image labels. Across three distinct medical imaging modalities—brain MRI, chest CT, and lung X-ray—MWM consistently outperforms established MIM baselines such as SparK, delivering gains of up to +3.1 percentage points in classification accuracy and, on brain-tumor transfer, +1.3 in box average precision (BoxAP) and +1.1 in mask average precision (MaskAP). Critically, it realizes these improvements while operating at a substantially lower overall masking ratio (e.g., 40% versus 70%), decoupling representation quality from the high masking ratios that MIM has conventionally required. Together, these results establish controllable, text-driven masking as an effective and annotation-free route to semantically aligned self-supervised learning for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute the foundational tasks of computer vision, and deep learning is the core technology driving progress across all three. That progress, however, is predicated on access to large-scale annotated corpora. In specialized domains such as medical image analysis, high-quality annotations are scarce, costly, and dependent on expert readers, and this annotation bottleneck constitutes a decisive barrier to the deployment of deep learning at scale.

Self-supervised learning (SSL) confronts this bottleneck directly, pretraining on unlabeled data to learn transferable representations without manual annotation [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among competing SSL paradigms, masked image modeling (MIM) has emerged as a leading approach: by occluding portions of an image and requiring the encoder to reconstruct the missing content, MIM compels the network to infer structural and semantic regularities from the visible context. This objective attains strong performance across a

broad range of downstream tasks and transfers successfully to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

A central weakness, however, is intrinsic to the objective as commonly instantiated: prevailing MIM approaches—including masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)]—adopt random masking that is, by construction, agnostic to the regions of interest that govern downstream tasks. This task-agnostic masking incurs two coupled limitations. (1) The learned representations are prone to semantic misalignment with the downstream objective, which erodes transfer performance. (2) Because random masking can raise the probability of occluding informative content only by occluding more of the image, existing methods resort to extreme masking ratios (e.g., 75% in MAE), inflating reconstruction difficulty and, with it, the demand for large-scale data and compute. Natural language stands in instructive contrast: its compact structure and high semantic density support effective masked pretraining even at low masking ratios [Devlin et al. (2019)]. The reliance on high ratios in image-based MIM thus traces directly to two entangled causes—inherent pixel-level redundancy and the absence of semantic-aware masking—which jointly prevent the model from concentrating on the regions that actually matter.

These deficiencies are especially pronounced in medical imaging, where two properties compound them. First, medical images exhibit acute semantic sparsity: diagnostically relevant information is typically confined to small, localized regions such as lesions or organs, whereas the majority of the field of view is semantically redundant background. Second, the domain spans heterogeneous modalities—MRI, CT, and X-ray—across which the shape, scale, and spatial distribution of task-relevant regions vary substantially from one task to the next. In combination, these properties impose unusually stringent demands on both the efficiency and the generalizability of any masking strategy, and they are precisely the demands that random masking is ill-equipped to meet.

We resolve these challenges with **Mask What Matters (MWM)**, a controllable, text-guided masking framework. Given a user-specified, open-vocabulary description or phrase, a frozen pretrained vision-language model supplies zero-shot localization cues, which MWM consolidates into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—allocating higher ratios to semantically salient structures such as lesions and organs and lower ratios to background—so that downstream semantics are injected directly into the pretraining signal, and, crucially, without recourse to per-image reports or labels. The masking prior is thus supplied by language rather than by supervision or reconstruction difficulty, which is what allows a single mechanism to remain both task-aware and annotation-free.

In summary, this work makes the following contributions:

- **A controllable text-guided masking framework for medical imaging.** We introduce, to our knowledge, the first framework that steers masked image modeling through open-vocabulary text. MWM couples a frozen vision-language model (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from natural-language prompts, and it turns the resulting localization into a masking prior via differentiated ROI-versus-background allocation—the point at which downstream semantics enter pretraining and the semantic misalignment inherent to random masking is removed.
- **An annotation-free, backbone-agnostic design.** MWM dispenses with per-image reports and labels entirely, deriving its guidance solely from prompts and frozen off-the-shelf models. Its visible-patch interface integrates into both ViT- and ConvNet-based MIM pipelines under a single unified protocol, rendering the localization-then-masking mechanism portable across architectures rather than tied to a particular backbone.
- **Consistent gains across modalities, tasks, and masking ratios.** On brain MRI, chest CT, and lung X-ray, MWM surpasses strong MIM baselines such as SparK and AnatoMask, improving image classification, brain-tumor detection, and instance segmentation alike—and it sustains this advantage at markedly lower masking ratios (e.g., 40% versus 70%), turning the semantic prior into a concurrent gain in representation quality and pretraining efficiency.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling originated in natural language processing with the bidirectional encoder representations from transformers (BERT) (Devlin et al., 2019), which learns contextual representations by masking tokens and predicting them during pretraining. Building on this principle, He et al. (2022) transferred MIM to computer vision, occluding portions of an image at pretraining time and tasking the model with reconstructing the missing regions. The paradigm generalizes strongly across both natural and medical image domains (Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023). Representative instantiations include MAE (He et al., 2022), MedMAE (Gupta et al., 2025; Xiao et al., 2023), and SparK (Tian et al., 2023): MAE employs high-ratio random masking to drive ViTs toward high-level semantic representations; MedMAE adapts this recipe to medical images through domain-specific modifications; and SparK ports the strategy to convolutional backbones, exploiting their intrinsic multi-scale features to strengthen pretraining. A common thread nonetheless unites them: the masking distribution is decoupled from task semantics.

This decoupling is the crux of the limitation. Because conventional MIM masks randomly or structure-agnostically, it disregards the semantic requirements of downstream tasks and thereby leads the model to expend capacity reconstructing irrelevant regions during pretraining, diminishing transfer effectiveness. Recent work has consequently turned to region-aware masking as a means of restoring semantic alignment, and two principal directions have emerged.

The first line of work introduces loss-driven adaptive masking. Methods such as hard patches mining (HPM) (Wang et al., 2023a) and its extension AnatoMask (Li et al., 2024) dynamically analyze per-region reconstruction errors and assign higher masking probabilities to regions with larger losses, on the premise that reconstruction difficulty correlates with semantic value. That premise is fragile: difficulty and task relevance are not equivalent, so the strategy can steer the model toward hard-to-reconstruct yet task-irrelevant structures. In brain magnetic resonance imaging (MRI), for instance, intricate structures such as the spinal cord or the eyeballs may absorb disproportionate attention despite being irrelevant to downstream objectives such as tumor detection.

The second line of work enlists external perceptual modules to guide masking. FocusMAE (Basu et al., 2024), for example, leverages pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although effective on its target tasks, this strategy hinges on supervised training of RPNs over domain-specific annotated data, reintroducing the very annotation dependency that SSL seeks to eliminate and, in doing so, curtailing cross-task generalization.

Both directions therefore leave the same gap unfilled: a masking mechanism that is at once flexible, semantically aligned, and generalizable without demanding additional supervision. Natural language—an interpretable and readily available source of guidance—offers a compelling means of closing this gap, steering masked modeling in a manner that is task-aware yet annotation-free.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid maturation of vision-language alignment models has made it increasingly feasible to perceive and localize task-relevant regions under zero-shot, open-world conditions (Radford et al., 2021). Foundational approaches such as the Contrastive Language-Image Pretraining (CLIP) family induce a unified image-text embedding space through large-scale joint pretraining on image-text pairs, and they now serve as standard tools for zero-shot and open-set visual tasks. Within medical imaging, domain-specialized extensions such as BiomedCLIP (Zhang et al., 2023) sharpen this cross-modal understanding and deliver robust performance across a diverse range of downstream applications. The deployment of these models, however, has remained largely confined to supervised settings: existing studies predominantly target downstream applications such as object detection and semantic segmentation (Zhong et al., 2022; Aleem et al., 2024; Koleilat et al., 2024), and none, to our knowledge, has harnessed their open-vocabulary, zero-shot capabilities to guide masked image modeling itself. It is precisely this unexploited opportunity that our approach sets out to realize.

Two observations crystallize the opening. First, recent region-aware MIM approaches—whether reconstruction-loss-driven dynamic masking or masking guided by external perception modules—improve semantic focus but remain hampered by persistent shortcomings: dependence on

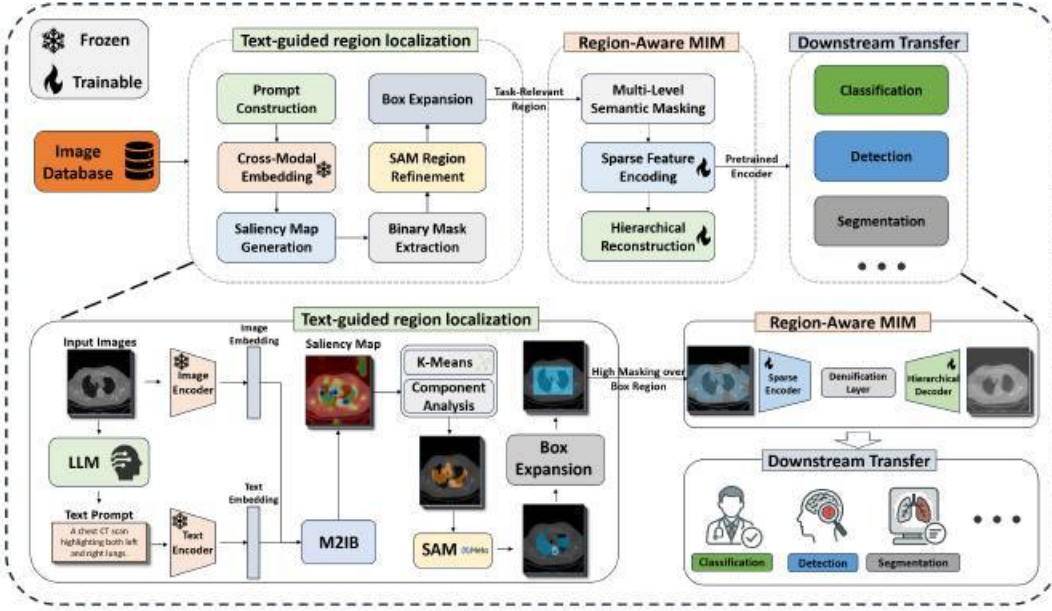


Figure 1: Overview of the MWM framework. The top panel depicts the end-to-end pipeline in three stages: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel details the mechanism by which an open-vocabulary prompt drives region localization and, in turn, the differentiated masking distribution that conditions pretraining.

additional supervision, residual misalignment between masked regions and task semantics, and limited flexibility. Second, despite the demonstrated strength of vision–language models on downstream tasks, natural-language prompts have not been brought to bear on the MIM pretraining phase to supply task-aware masking guidance. These observations converge on the direction this work pursues: leveraging vision–language semantic guidance to render masked image modeling at once semantically aligned, efficient, and generalizable across the medical modalities we study.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is organized into two tightly coupled stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage employs a frozen vision–language model to identify task-relevant regions from an open-vocabulary prompt; the second consumes those regions and imposes differentiated masking ratios across them, thereby conditioning the reconstruction objective on downstream semantics. The two stages are decoupled in computation yet unified in purpose: localization determines *where* the learning signal is concentrated, and masking determines *how strongly*. This separation is what lets a single semantic prior—the prompt—govern the entire pretraining signal.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a deterministic, multi-stage pipeline in which each stage refines the spatial estimate produced by its predecessor. The pipeline begins with LLM-based prompt generation and encodes image and prompt with BiomedCLIP to obtain aligned cross-modal embeddings. From these embeddings it derives a saliency map via the Multi-Modal Information Bottleneck (M2IB), binarizes the map with K-Means clustering, and refines the resulting region with the Segment Anything Model (SAM). The refined mask is finally converted into a bounding box endowed with a controllable expansion margin, which serves as a robust, noise-tolerant spatial guide for the subsequent masked modeling stage. Each component is detailed below.

Prompt Generation. The textual prompt is decisive for the quality of region localization. To strengthen the semantic guidance supplied to the vision–language model, we query a large language model (LLM) with a structured template of the form “Describe the typical visual characteristics of a [category] in a [modality] image”, in which *category* and *modality* are instantiated per task. As a lightweight alternative, we also admit bare category phrases as inputs. The two prompt styles trade descriptive richness against specificity; we compare them systematically in the experimental section and find, as reported there, that the more accurate style is modality-dependent.

Cross-Modal Embedding. Given the generated prompt, a frozen BiomedCLIP model [Zhang et al. (2023)] encodes the medical image and its prompt into a shared semantic space in which cross-modal similarity is well defined. For an image I and a prompt T , the embeddings are computed as

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These aligned embeddings constitute the semantic foundation on which the saliency estimation of the subsequent stage operates.

Saliency Map Generation. To translate the aligned embeddings into a spatial estimate, we employ a cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)], which localizes task-relevant visual regions by maximizing image–text agreement while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns to each spatial location an importance weight proportional to its semantic relevance to the prompt. M2IB obtains this map by optimizing the objective

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings produced by BiomedCLIP, and the coefficient γ governs the trade-off between preserving cross-modal relevance (the first term) and discarding task-irrelevant visual information (the second term).

Binary Mask Extraction. The continuous saliency map is next binarized into a preliminary foreground mask by unsupervised K-Means clustering [Lloyd (1982)]: the saliency values are partitioned into two clusters and the higher-valued cluster is designated as foreground—an adaptive, threshold-free criterion that obviates manual tuning. To suppress spurious activations, connected-component analysis is then applied to the binary map, retaining only the largest foreground components. This step discards noisy edge fragments and concentrates the mask on the core semantic structures.

Region Refinement. To sharpen spatial accuracy beyond the coarse cluster boundaries, SAM is incorporated as a refinement module [Kirillov et al. (2023)]. For each retained connected region $c_i \in \mathcal{C}^*$, we compute its minimal enclosing bounding box and pass it as the visual prompt to SAM.

Conditioned on the original image I and the set of bounding-box prompts, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} recovers spatial structure with markedly higher boundary fidelity than the coarse cluster.

Box-Based Region Expansion. Residual boundary noise in the SAM mask can still misdirect the model’s focus during representation learning. To render the region guidance robust to such noise, the refined SAM mask is converted into a bounding box endowed with a controllable expansion margin; this expanded box constitutes the final region of interest (ROI) that conditions the downstream masked modeling stage. The margin exposes a single, interpretable control over how conservatively the ROI encloses the localized structure, and it is by tolerating—rather than propagating—residual localization error that the pipeline remains stable under imperfect upstream estimates.

3.3 REGION-AWARE MASKED IMAGE MODELING

Conditioned on the ROI produced by the localization module of Section 3.2, MWM performs masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that assigns differentiated masking ratios according to region importance, (2) a sparse encoder that processes only the visible (unmasked) patches, and (3) a hierarchical decoder that reconstructs the full image from the resulting sparse multi-scale features. The three stages compose into a single reconstruction objective in which the masking distribution—not the architecture—carries the semantic

prior; this design choice is what allows the same backbone to inherit task awareness solely through where it is asked to reconstruct.

Multi-Level Semantic Masking. MWM couples the masking distribution directly to the semantic regions delivered by text-driven localization. Patches falling within the prompt-highlighted ROI—regions associated with tumors or organs—are masked at a high ratio, compelling the model to reconstruct, and thereby to encode, the task-relevant semantic features. Background patches, by contrast, are masked lightly or left visible, curtailing redundant reconstruction effort and its attendant computation. This differentiated allocation is precisely what allows MWM to place a strong learning signal on sparse, diagnostically critical structures without inflating the global masking ratio. Figure 2 contrasts the resulting masking pattern with conventional random masking.

Sparse Feature Encoding. The visible patches are gathered into a sparse image and encoded following the Spark design Tian et al. (2023):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates its visibility, Φ_{sparse} is the sparse encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are the resulting multi-scale features. Although we instantiate MWM with convolutional backbones in the experiments that follow, the same visible-patch interface applies unchanged to ViTs via token indexing; it is this indexing abstraction, rather than any backbone-specific mechanism, that renders the masking strategy architecture-agnostic.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in top-down fashion. At each scale l , masked locations are populated with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , which is then upsampled and fused with the features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The model is trained with a mean-squared-error objective computed exclusively over the masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction. Because the loss is confined to masked locations and masking is concentrated on the text-localized ROI, the training signal is, by construction, directed toward the diagnostically relevant regions—closing the loop between the prompt, the localized ROI, and the gradient that shapes the representation.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining is conducted on three datasets that together span three distinct imaging modalities, so that the evaluation exercises MWM across markedly different anatomical and appearance statistics rather than a single acquisition regime.

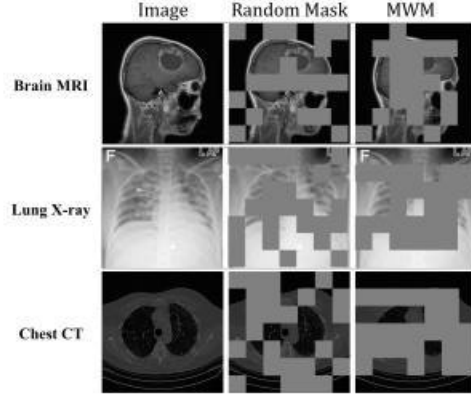


Figure 2: Comparison of masking strategies. Gray blocks denote masked regions: whereas random masking distributes occlusion uniformly, MWM concentrates masking on the text-localized ROI while leaving the redundant background largely visible.

(1) *Brain MRI* comprises approximately 17,000 images drawn from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering a range of tumor types and anatomical structures. (2) *Lung X-ray* comprises approximately 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung-disease dataset [Konya (2020)], accompanied by fibrotic-lung segmentation masks from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation spans classification and detection tasks over three further datasets, deliberately chosen so that each pretraining modality is transferred to an *independent* downstream benchmark rather than re-evaluated on its own pretraining images. (1) *Brain Tumor MRI* [Feltrin (2023)] contains 4,479 MRI images spanning 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermay et al. (2018)] contains 5,863 pediatric chest X-rays labeled as normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] contains CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

To ensure a fair comparison, all self-supervised methods are built on backbones of comparable scale: convolution-based methods adopt ResNet-50 [He et al. (2016)] and transformer-based methods adopt ViT-S [Dosovitskiy et al. (2020)]. All models operate at an input resolution of 224×224 and are initialized from ImageNet-pretrained weights, and downstream fine-tuning retains the same encoder and input resolution as pretraining. Pretraining uses an initial learning rate of 2×10^{-4} . All experiments are conducted on NVIDIA Tesla A100 GPUs with 80 GB of VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless stated otherwise, non-target variables—for instance, prompt type or SAM usage—are held fixed at their empirically preferred settings, so that each reported comparison isolates the single factor under study; the prompt style used per modality is the one identified as more accurate in the localization analysis of Section 4.3, and no downstream test labels enter this selection.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first establish that the text-guided localization module reliably identifies task-relevant regions across all three imaging modalities—brain MRI, lung X-ray, and chest CT—and then quantify the separate contributions of prompt type and SAM refinement. We compare the predicted bounding boxes against expert-annotated segmentation masks, reporting occlusion precision ($|R_p \cap R_{gt}| / |R_p|$) and occlusion recall ($|R_p \cap R_{gt}| / |R_{gt}|$), where R_p denotes the predicted region and R_{gt} the ground-truth annotation. These available masks index the same diagnostic targets—tumors and lung fields—that the prompts describe, so recall measures the coverage that matters for downstream masking.

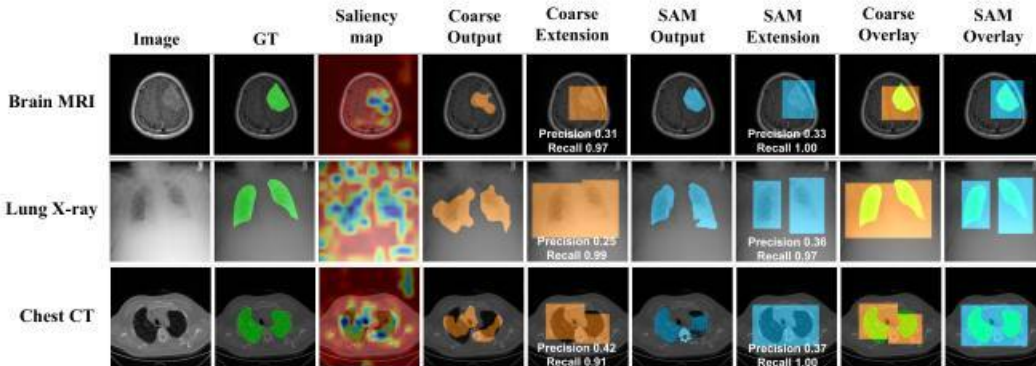


Figure 3: Representative text-guided region localization results across the three modalities, illustrating that open-vocabulary prompts alone recover the diagnostically relevant regions—tumors in brain MRI, lung fields in chest CT, and pulmonary regions in lung X-ray.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Text-guided localization performance, measured as occlusion precision and recall against expert segmentation masks, across prompt types and refinement strategies on brain MRI, chest CT, and lung X-ray. *Coarse Ext.* denotes bounding-box extension of the coarse cluster region; + *SAM Ext.* denotes further refinement with SAM. The best value in each column is shown in bold.

As reported in Table 1, text-guided localization with an effective prompt attains consistently high occlusion recall across all three datasets—on the order of 0.82–0.94—while maintaining competitive precision. The framework thus recovers task-relevant regions from textual descriptions alone, and it does so across pronounced modality-specific anatomical variation; Figure 3 provides qualitative corroboration. Precision is expectedly lower than recall by design, since the expanded bounding box is intended to enclose the target with margin rather than to segment it exactly.

Effect of Prompt Type on Region Localization. To characterize the role of language design, we compare the two prompt types under otherwise identical conditions. As Table 1 shows, descriptive sentence prompts are superior on brain MRI and chest CT, indicating that richer semantics aid the localization of small, spatially concentrated structures. On lung radiographs, by contrast, concise phrases achieve higher recall (0.82 vs. 0.66): the broad anatomical extent of the lungs renders general terms sufficient, so the added specificity of a full sentence confers no benefit and can even narrow coverage. The most accurate prompt granularity is therefore modality-dependent—a relationship we exploit, rather than tune away, in the downstream experiments.

Effect of SAM Refinement on Region Localization. We next isolate the contribution of spatial refinement by applying SAM to the coarse masks. As Table 1 shows, SAM refinement consistently improves precision without sacrificing recall: on brain MRI, precision rises from 0.22 to 0.30 while recall remains above 0.94. SAM thus tightens the localized region around the true structure, sharpening the ROI that conditions masking while preserving the coverage established by the prompt.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We now assess the representation quality and generalization of MWM against representative self-supervised learning (SSL) methods drawn from both the reconstruction-based (MIM) and contrastive learning (CL) paradigms. To attribute any measured difference to the method rather than to incidental confounds, all baselines are pretrained and evaluated on identical datasets under identical training settings. We note that MWM additionally draws on frozen off-the-shelf models (BiomedCLIP and SAM) that the purely image-based SSL baselines do not; the ablations in Section 4.5 isolate the contribution of the masking mechanism itself by comparing text-guided against random masking within an otherwise fixed pipeline, so that the reported gains are not merely an artifact of these external components.

Image Classification. We examine classification performance on the three downstream medical imaging datasets under two complementary protocols: (1) *full fine-tuning*, in which the entire model is updated, measuring end-to-end transfer; and (2) *linear probing*, in which the encoder is frozen and only a linear classifier is trained atop it, isolating the intrinsic quality of the pretrained representation. The two protocols probe complementary facets of transfer, and MWM leads under both.

- **Full fine-tuning.** Table 2 reports classification accuracy for MWM against representative self-supervised methods spanning both paradigms: MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and BYOL [Grill et al. (2020)]. MWM attains the highest accuracy on every one of the three modali-

ties, evidence of consistent generalization across modalities rather than a modality-specific effect. The margin is largest on chest CT, where MWM reaches 97.5% and surpasses the strongest MIM baselines by clear margins of +3.1 over SparK and +1.7 over AnatoMask; on brain MRI and lung X-ray the gains over SparK are smaller but uniformly positive (+0.6 and +1.3, respectively). Chest CT is precisely the regime that text-guided masking is designed to exploit—one in which semantically critical structures such as the lung lobes are spatially sparse against an expansive redundant background—so that concentrating the learning signal there, rather than spending capacity reconstructing background, yields the greatest benefit.

- **Linear probing.** To interrogate representation quality directly, we freeze the encoder and train only a linear classifier, comparing MWM against the two strongest MIM baselines, SparK and AnatoMask. As Table 3 shows, MWM outperforms both on all three datasets, with pronounced margins over AnatoMask of +3.2 points on lung X-ray (88.5 vs. 85.3) and +3.1 on chest CT (67.0 vs. 63.9). Because the backbone is never updated in this setting, the advantage is attributable to the pretrained features themselves: text-driven pretraining yields representations that are discriminative even absent any fine-tuning.

Object Detection and Instance Segmentation. To verify that the learned representations transfer beyond classification to dense prediction, we evaluate brain-tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As Table 4 shows, MWM attains the highest AP^{box} (46.9%), surpassing SparK by +1.3, and the best AP^{box}_{75} (53.5%); on segmentation it improves AP^{mask} over SparK by +1.1 (45.9 vs. 44.8) and ties AnatoMask for the best AP^{mask}_{75} (54.1%), remaining within 0.3 of AnatoMask on AP^{mask} . MWM thus transfers effectively to both object-level localization and pixel-wise segmentation, carrying its advantage from image-level classification into spatially structured prediction.

Taken together, these experiments indicate that text-driven masking during pretraining yields more informative and more transferable representations. The gains recur across classification, detection, and instance segmentation, which establishes that the benefit of semantic masking guidance is task-general within the medical settings evaluated rather than confined to any single downstream objective.

4.5 ABLATION STUDY

We conduct controlled ablations that isolate the individual contribution of each design choice in MWM—prompt design, SAM-based region refinement, and masking ratio—to downstream classification performance. Because each variant is evaluated within an otherwise fixed pipeline, these ablations serve a second purpose: they disentangle the effect of the masking mechanism from that of the external models it employs.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As reported in Table 5, descriptive sentence prompts surpass short phrases on brain MRI and chest CT, whereas phrases prevail on lung X-ray. This pattern mirrors the localization results of Section

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Full fine-tuning classification accuracy (%) on the three downstream datasets. MIM denotes Masked Image Modeling and CL denotes Contrastive Learning. All methods share backbone scale, datasets, and training settings. MWM attains the highest accuracy on every modality, with the largest margin on chest CT (+3.1 over SparK); the best value in each column is shown in bold.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear-probing accuracy (%) on the three downstream datasets with a frozen encoder, which isolates the intrinsic quality of the pretrained representation. Compared against the two strongest MIM baselines, MWM leads on every modality; the best value in each column is shown in bold.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Brain-tumor detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) on the BR35H dataset (%), comparing MWM against the two strongest MIM baselines. MWM leads on both detection metrics and on AP_{75}^{mask} , and is a close second to AnatoMask on AP^{mask} ; the best value in each column is shown in bold.

4.3: for each modality, the prompt style that localizes more accurately is the one that yields the stronger downstream accuracy, which links localization quality to transfer performance rather than leaving the two merely correlated. More decisively, both prompt types—irrespective of granularity—outperform the No-Prompt (random-masking) baseline on every dataset. Since prompt wording varies while the improvement persists, the operative factor is text-guided masking itself rather than any particular phrasing: aligning the pretraining signal with downstream semantics is what drives the gains, and it does so even when the prompt is only approximately specified. The SAM ablation completes the picture: incorporating region refinement during pretraining consistently improves downstream classification, corroborating the value of sharpening the text-guided ROI, while the No-Prompt comparison confirms that the external models contribute through the masking prior they induce rather than as a standalone advantage.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM reaches 96.8% classification accuracy at a masking ratio of only 40%—well below the roughly 70% conventionally required in masked image modeling—and, despite this reduced ratio, exceeds SparK’s best result at 70% masking by +2.4 points in absolute accuracy. To exclude the possibility that this comparison merely reflects a mis-configured baseline, we sweep SparK across 40%, 50%, 60%, and 70% masking ratios. SparK peaks at 70%, which confirms that MWM is measured against SparK’s own optimum and that its advantage is therefore not an artifact of baseline tuning. These findings challenge the prevailing assumption that high masking ratios are inherently optimal for masked image modeling. What governs representation quality is not how much is masked but where: by preserving task-relevant content and eliminating redundant masking, semantically guided strategies such as MWM attain stronger downstream performance at substantially lower masking ratios. This reduced ratio lowers the reconstruction workload during pretraining; we note, however, that

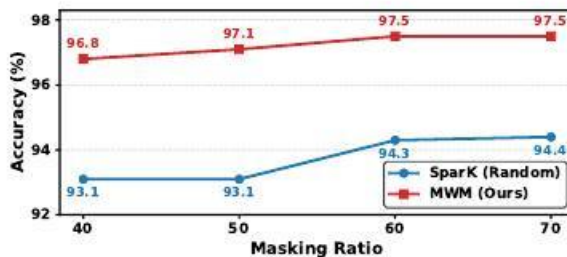


Figure 4: Chest CT classification accuracy as a function of masking strategy and ratio. MWM peaks at a 40% masking ratio and exceeds SparK’s own best (attained at 70% under a matched sweep), demonstrating that semantic guidance decouples accuracy from the high masking ratios MIM has conventionally required.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under varying prompt types and region-refinement strategies across the three imaging modalities. *No Prompt* denotes random masking without text-guided localization and serves as the baseline within an otherwise identical pipeline; *w/* and *w/o* denote with and without SAM, respectively. Every text-guided variant exceeds the No-Prompt baseline on all modalities, isolating semantic masking—rather than the external models per se—as the operative factor; the best value in each column is shown in bold.

MWM incurs a one-time localization cost—BiomedCLIP inference, clustering, and SAM refinement—that random masking does not, so the reduced ratio is best read as evidence that semantic guidance decouples accuracy from ratio rather than as a claim of net end-to-end speedup.

5 CONCLUSION AND LIMITATIONS

This work introduced **Mask What Matters (MWM)**, a self-supervised pretraining framework that unifies text-guided semantic localization with region-aware masking. By using open-vocabulary natural-language prompts to identify task-relevant regions and applying differentiated masking to them, MWM realizes semantically aligned representation learning without any per-image annotation, drawing its guidance entirely from prompts and frozen off-the-shelf models. Its principal contribution lies not in any single component but in their integration: the use of prompt-derived localization as a controllable masking prior that couples pretraining to downstream semantics.

Our experiments substantiate the approach along three fronts: (1) natural-language prompts reliably localize semantic regions across all three modalities, establishing that text-driven masking is feasible without task-specific supervision; (2) MWM consistently surpasses strong baselines such as SparK and AnatoMask in classification across brain MRI, chest CT, and lung X-ray, and it delivers further gains in brain-tumor detection and instance segmentation, indicating that the benefit is task-general across the settings evaluated rather than tied to a single objective; (3) MWM sustains this advantage even at markedly lower masking ratios (e.g., 40% versus 70%), showing that semantic guidance decouples representation quality from the high masking ratios MIM has conventionally required.

Limitations. Several limitations delimit the present study and define its immediate follow-up. First, the reported downstream margins—particularly under full fine-tuning—are in some cases modest, and we have not accompanied them with variance estimates across seeds or splits; quantifying this variability is a priority for establishing the robustness of the smaller gains. Second, although text prompts furnish flexible guidance, we have not systematically characterized MWM’s robustness to prompt style, vocabulary, or noise, nor to failures of the external vision–language model and segmentation model on diseases or modalities poorly represented in their pretraining data. Third, the localization pipeline comprises several stages—prompt wording, saliency estimation, K-Means binarization, connected-component filtering, and SAM-based refinement with box expansion—and a systematic analysis of its failure cases and sensitivity to these design choices remains open. Finally, the evaluated tasks all involve comparatively structured anatomical targets; extension to diffuse disease, multi-organ settings, or weakly localized labels, where suitable prompts may be ambiguous, is not yet demonstrated. The first three concerns bear on the prompt interface and its supporting models rather than on the masking mechanism itself, and each points to a concrete direction for strengthening the framework.

Looking ahead, text-driven, region-aware masking offers a principled route toward more semantically grounded self-supervised learning; characterizing prompt robustness and computational overhead, and extending MWM beyond the medical modalities studied here, are natural next steps toward broader vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.